UPSA
UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA
Scholarship with Professionalism

BSIT 401 Network and Systems Administration

Chapter 3: MANAGING SERVERS AND WORKSTATIONS

Lecturer

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1.0 Week 3 Overview

Servers and workstations are the operational endpoints on which organizational applications, data, authentication, file services, databases, web services and user productivity depend.

A network and systems administrator therefore needs to understand not only what these systems are, but also how to install, configure, update, monitor, troubleshoot and remotely administer them throughout their lifecycle.

Week 3 builds on the resource-inspection skills from Week 2 and shifts the focus from hardware resources to operational system administration.

1.1 Learning Outcomes

- Explain the roles and major types of servers and workstations in organisational IT infrastructure.
- Distinguish server-class and workstation-class administration requirements.
- Explain the operating-system lifecycle from planning and installation through operation, maintenance, upgrade and retirement.
- Inspect and manage Linux processes and services using standard administrative commands.
- Use package-management tools to refresh repositories, inspect packages and perform safe updates.
- Identify failed services and diagnose common server/workstation readiness problems using evidence.
- Explain secure remote administration using SSH and compare it with common Windows remote-management methods.
- Prepare and apply a practical server-readiness checklist before placing a system into service.
- Document administrative actions, evidence, changes and verification results professionally.

1.2 Key Terminology

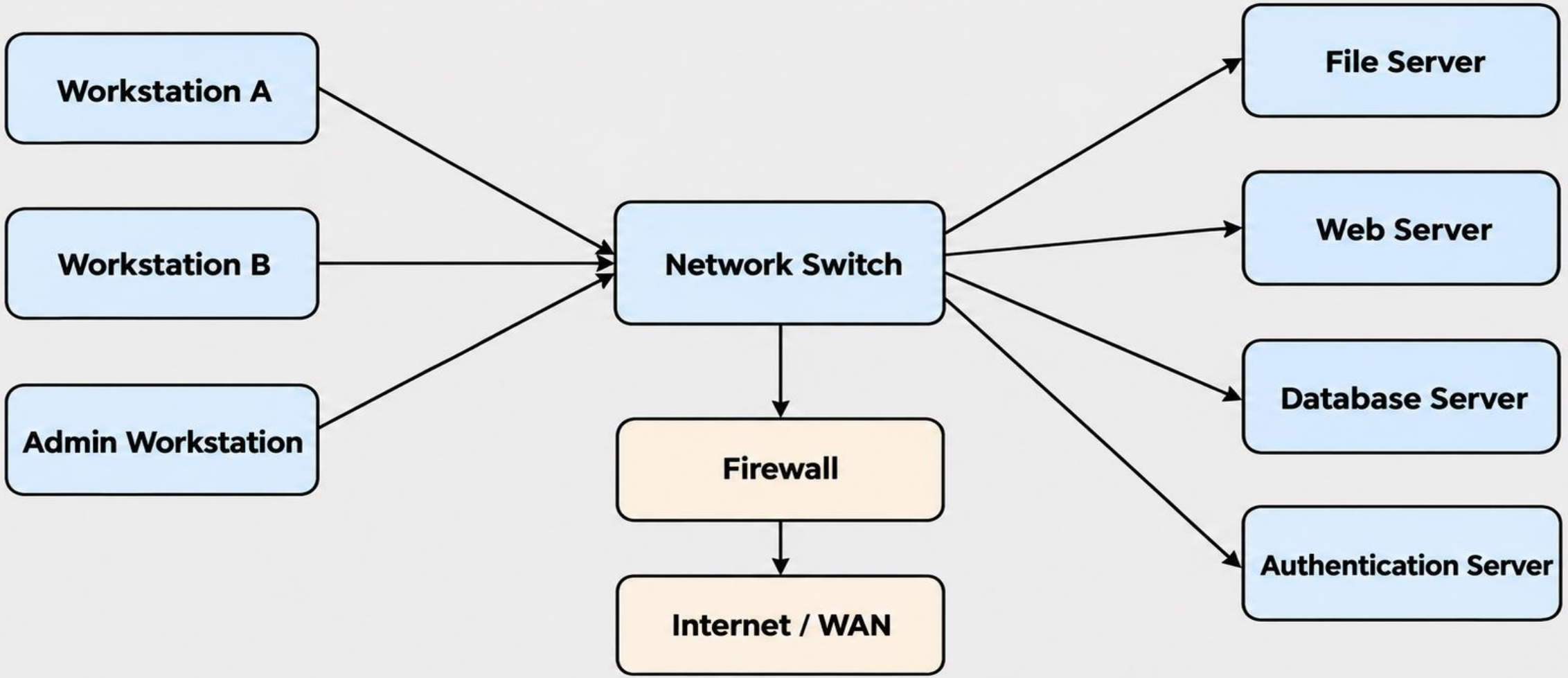
Term	Meaning in Administration
Server	A computer system or software instance that provides services, resources or data to other systems or users.
Workstation	A client-oriented computer used by an individual for professional or general-purpose computing tasks.
Service/Daemon	A background program that provides a system or network function, often managed by a service manager such as systemd.
Process	A running instance of a program with a process identifier (PID), state and resource usage.
Package	A distributable software unit containing program files, metadata and dependency information.
Repository	A trusted software source from which packages and updates are obtained.
Patch	A software change intended to fix bugs, vulnerabilities or stability issues.
Remote administration	Managing a computer over a network using secure tools such as SSH, RDP or PowerShell remoting.
Server readiness	The condition in which a server is correctly installed, configured, updated, secured, monitored and verified for its intended role.
Lifecycle	The stages through which an operating system passes from planning and installation to operation, maintenance, upgrade and retirement.

2.0 Servers and Workstations in Organizational Infrastructure

A server is designed primarily to provide a service to other systems, applications or users, while a workstation is primarily used by an individual user to consume services, create content, run applications and access organizational resources.

The distinction is functional rather than purely physical: a powerful desktop can provide a server service, and a virtual machine can function as a server even though it has no dedicated physical chassis.

Typical Organisational Server-Workstation Architecture



1. WEEK OVERVIEW

This week focuses on the administration of servers and workstations, operating system lifecycle, package management, processes, services, updates, remote administration and basic server readiness. Students will learn how to install, configure, monitor, and maintain servers and workstations to ensure reliable and secure operation in an organisational environment.

2. LEARNING OUTCOMES

By the end of this week, you should be able to:

1. Explain the roles and types of servers and workstations in an organisation.
2. Install, configure and manage operating systems.
3. Use package managers to install and update software.
4. Manage services and processes on servers and workstations.
5. Perform system updates and apply patches.
6. Configure and use remote administration tools (SSH, RDP, VNC).
7. Assess basic server readiness for production use.

3. SERVERS AND WORKSTATIONS: AN OVERVIEW

- A server is a computer system that provides services, resources or data to other computers across a network.
- A workstation is a client computer used by an individual for tasks such as application development, data processing and accessing network resources.
- Both require proper administration, configuration, monitoring and maintenance to ensure availability, performance and security.

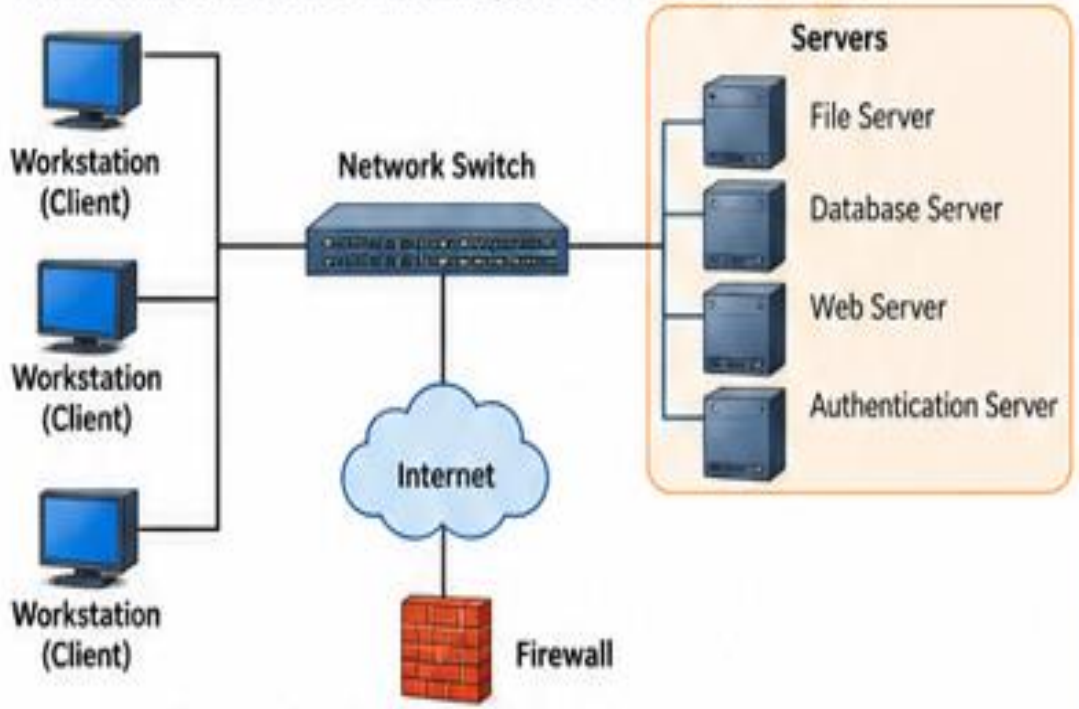


Figure 1: Typical network with servers and workstations

4. TYPES OF SERVERS AND THEIR FUNCTIONS

Server Type	Function	Examples
File Server	Stores and manages files for users and applications.	Samba, Windows File Server
Web Server	Hosts websites and web applications.	Apache, Nginx, IIS
Database Server	Manages databases and provides data services.	MySQL, PostgreSQL, Microsoft SQL Server
Authentication Server	Manages user authentication and directory services.	Active Directory, LDAP, FreeIPA
DNS Server	Resolves domain names to IP addresses.	BIND, Windows DNS
DHCP Server	Automatically assigns IP addresses to clients.	ISC DHCP, Windows DHCP
Application Server	Runs business applications.	Tomcat, JBoss, WildFly
Mail Server	Handles email services.	Postfix, Exchange

2.1 Common Server Roles

Server Role	Primary Function	Typical Technologies / Examples
File server	Centralised storage and controlled file sharing	Samba, Windows Server file services, NFS
Web server	Hosts websites and web applications	Apache HTTP Server, Nginx, Microsoft IIS
Database server	Stores and processes structured organisational data	PostgreSQL, MySQL/MariaDB, SQL Server
Authentication / directory server	Central identity, authentication and directory services	Active Directory, LDAP, FreeIPA
DNS server	Translates domain names into IP addresses and stores DNS records	BIND, Windows DNS
DHCP server	Automatically provides IP addressing and related parameters	ISC Kea, dnsmasq, Windows DHCP
Application server	Runs enterprise/business applications	Tomcat, JBoss/WildFly, .NET application hosting
Mail server	Sends, receives and stores email	Postfix, Exchange, Dovecot
Monitoring server	Collects metrics, logs and alerts	Prometheus, Zabbix, Nagios, Grafana integrations
Backup server	Coordinates backup storage, retention and restoration	Bacula, Veeam, enterprise backup platforms

2.2 Server versus Workstation: Administrative Comparison

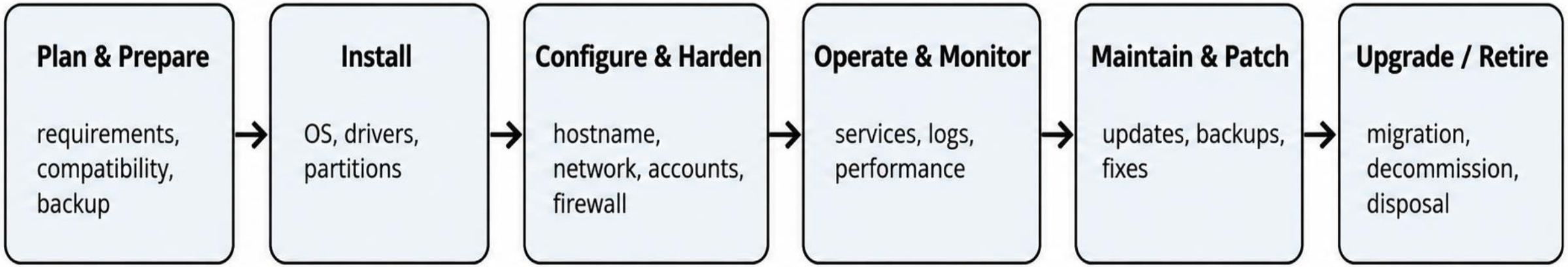
Dimension	Server	Workstation
Primary purpose	Provide shared service/resources	Support an individual user
Availability expectation	Often continuous or high availability	Usually tied to user working hours
Change control	Usually stricter; changes may affect many users	May be more flexible, but still controlled
Security exposure	Often network-facing and highly privileged	User-facing and exposed to user/application risks
Backup emphasis	Service/data/configuration recovery is critical	User files, profiles and application settings are important
Monitoring	Continuous service, capacity and log monitoring	Endpoint health, security and performance monitoring
Remote management	Common and often essential	Common in support and enterprise endpoint management
Hardware	May emphasise redundancy, ECC memory, RAID, multiple NICs	May emphasise interactive performance and peripherals

3.0 Operating-System Lifecycle

Operating-system administration is not a one-time installation task. Administrators manage systems through a lifecycle.

Each stage should be documented and controlled because decisions made early - such as partitioning, update policy, account structure or network configuration - affect reliability and security later.

Operating System Lifecycle for Servers and Workstations



Continuous documentation, security review, capacity planning and change control

3.1 Lifecycle Stages

Planning and preparation: Define the intended role, hardware requirements, operating-system edition, network requirements, storage layout, security requirements, backup plan, support lifecycle and change window.

Installation: Install the operating system, configure partitions/file systems, install required drivers and establish initial administrative access.

Initial configuration and hardening: Set hostname, time zone, network parameters, accounts, privilege controls, firewall rules, logging, updates and baseline configuration.

Operation: Provide the intended service while monitoring performance, service state, logs, capacity and security events.

Maintenance: Apply patches, update packages, rotate logs, verify backups, review accounts and permissions, replace certificates and correct faults.

Upgrade or migration: Move to a supported operating-system release or migrate applications/services to new systems with rollback and validation.

Retirement / decommissioning: Remove the service from production, migrate or archive data, revoke credentials, wipe sensitive storage where appropriate, update inventory and dispose of assets according to policy.

Administrative principle

A successful server installation is not the end of administration. The system must remain supportable, patchable, observable, recoverable and secure throughout its operational life.

4.0 Software Packages and Package Management

Linux distributions normally manage software through packages and trusted repositories. Package managers track versions, dependencies and installed files, making them safer and more consistent than manually copying application binaries. On Ubuntu/Debian systems, APT is the standard front-end used for common administrative package operations.

5. OPERATING SYSTEM LIFECYCLE

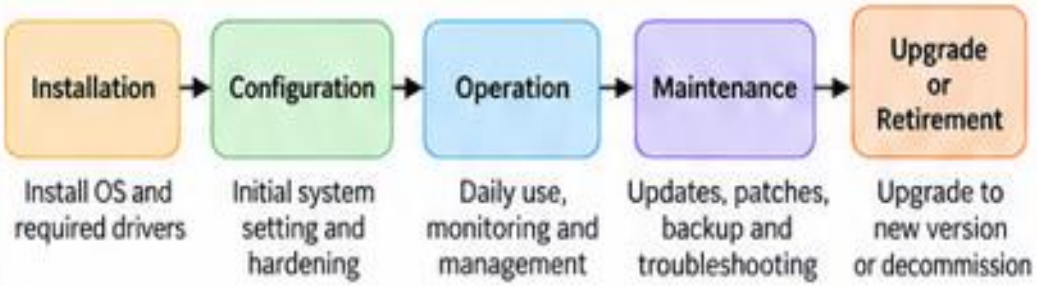


Figure 2: Operating system lifecycle

6. INSTALLING AN OPERATING SYSTEM (Ubunta Server Example)

1. Download the Ubuntu Server ISO from https://ubuntu.com/download/server
2. Create a bootable USB using Rufus or balenaEtcher.
3. Boot the server from USB and follow the installation wizard.
4. Configure hostname, user account, network settings and partitioning.
5. Complete the installation and reboot.
6. Verify installation using:

```
hostnamectl
cat /etc/os-release
```



Figure 3: Ubuntu Server installation screen

7. PACKAGE MANAGEMENT

- Package managers simplify the installation, update and removal of software.
- Common package managers:



Distribution	Package Manager	Example Commands
Ubuntu/Debian	apt	sudo apt update sudo apt install <package>
RHEL/CentOS	dnf / yum	sudo dnf install <package>
openSUSE	zypper	sudo zypper install <package>
Fedora	dnf	sudo dnf install <package>

8. MANAGING SERVICES AND PROCESSES

- Services (daemons) run in the background to provide specific functions.
- Use systemctl to manage services (systemd).

```
# Check service status
sudo systemctl status apache2

# Start, stop and enable a service
sudo systemctl start apache2
sudo systemctl stop apache2
sudo systemctl enable apache2
```

- Use ps, top or htop to view and manage processes.

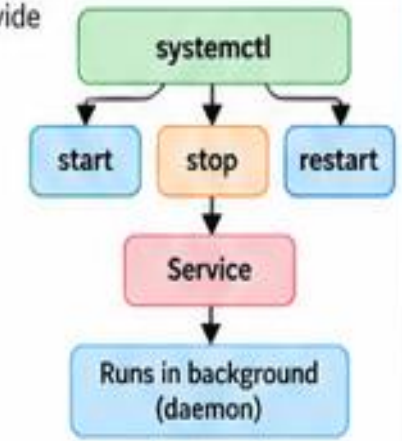


Figure 4: Service management with systemctl

9. SYSTEM UPDATES AND PATCH MANAGEMENT

- Regular updates fix bugs, patch security vulnerabilities and improve stability.
- Key commands (Ubuntu/Debian):

```
sudo apt update          # Update package lists
sudo apt upgrade         # Install available updates
sudo apt dist-upgrade    # Install updates, handle dependencies
```



Keep Systems Updated and Secure

10. REMOTE ADMINISTRATION

- SSH (Secure Shell) for Linux/Unix servers.
- RDP (Remote Desktop Protocol) for Windows workstations/servers.
- VNC for graphical remote access.
- Example SSH command:

```
ssh username@192.168.1.10
```



Figure 5: Remote administration tools

4.1 Core APT Concepts

Command	Purpose	Administrative Interpretation
sudo apt update	Refresh package indexes from configured repositories	Does not itself install most updates; it updates knowledge of what versions are available.
apt list --upgradable	List packages for which newer versions are available	Useful before changes to estimate update scope.
sudo apt upgrade	Install available upgrades without removing packages where possible	Routine update action; should still be performed with change awareness.
sudo apt install PACKAGE	Install a package and required dependencies	Use trusted repositories and verify package name and purpose.
sudo apt remove PACKAGE	Remove package binaries while usually retaining some configuration	Review dependencies and impact before removal.
sudo apt purge PACKAGE	Remove package and associated package-managed configuration	More destructive; use intentionally.
apt show PACKAGE	Display package metadata	Useful for version, dependencies and description.
dpkg -l | head	Inspect installed Debian packages	Provides lower-level package inventory evidence.

4.2 Update versus Upgrade versus Patch

In administration, the words are related but not identical. A package index update refreshes repository metadata; a package upgrade installs newer package versions; a patch is the actual software change that fixes a defect or vulnerability. A distribution upgrade may move the entire operating system to a newer release and therefore requires substantially more planning and testing.

5.0 Processes and Services

A process is a running program. A service is a long-running background function that normally exists to provide a system or network capability. On modern Linux distributions, systemd commonly manages services through systemctl. Administrators use service state together with process, socket and log evidence to determine whether an application is healthy.

5.1 Inspecting Running Services

```
systemctl list-units --type=service --state=running --no-pager
systemctl --failed --no-pager
systemctl status SERVICE --no-pager
```

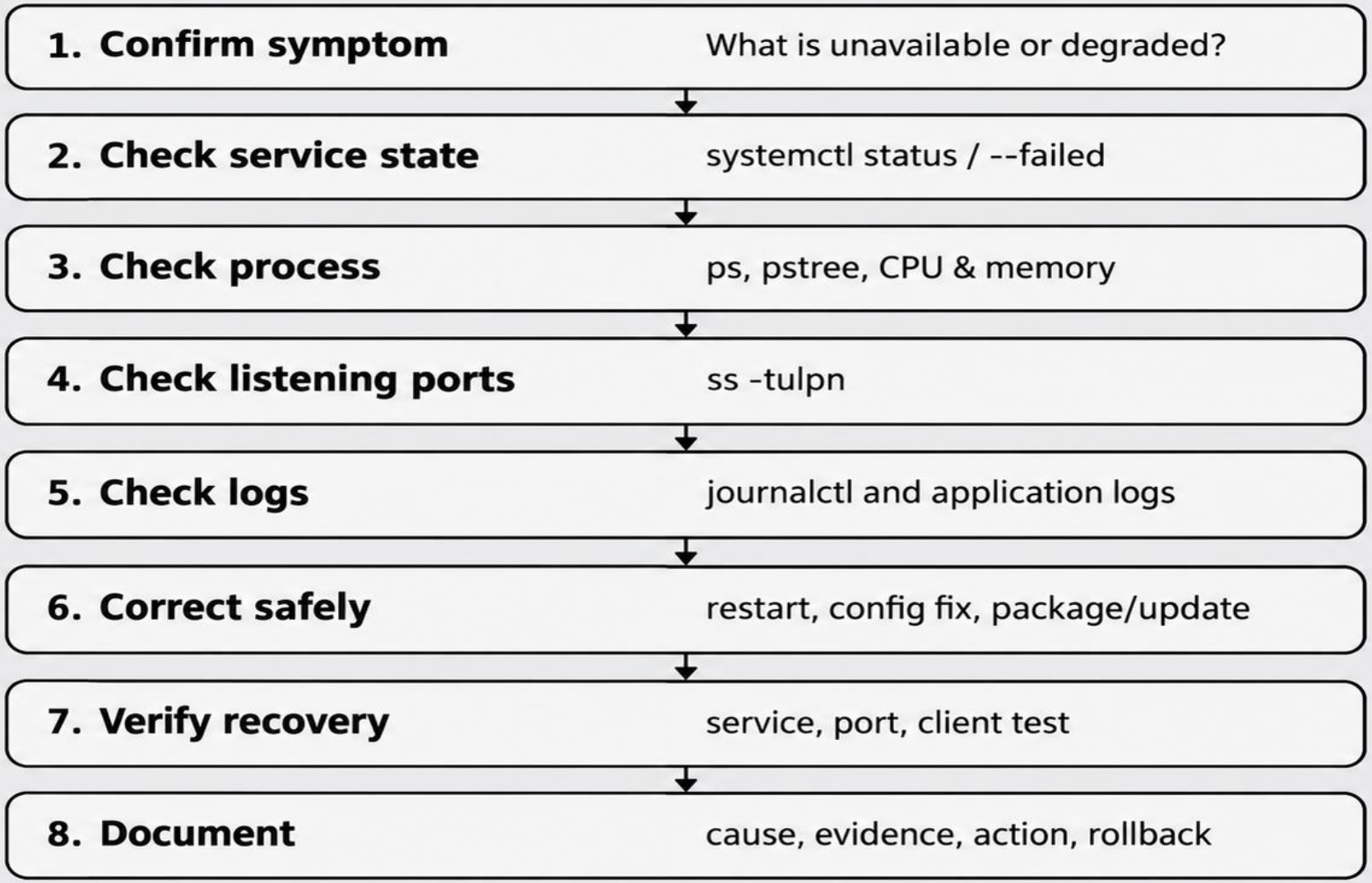
The first command lists active running services. The second focuses attention on units that **systemd** considers failed. The third provides status details for a selected service, including whether it is active, the main process ID, recent log messages and exit information when available.

5.2 Inspecting Processes

```
ps aux --sort=-%cpu | head
ps aux --sort=-%mem | head
pstree -p | head -n 30
```

Sorting by CPU or memory helps identify high resource consumers. **pstree** displays process ancestry, which is particularly helpful when a service launches worker processes or when an administrator needs to understand parent-child relationships.

Service Troubleshooting and Server Readiness Workflow



6.0 Remote Administration

Remote administration allows authorised administrators to manage systems without being physically present. For Linux servers, **SSH** is the most common secure command-line method. Windows environments commonly use **Remote Desktop Protocol (RDP),** Windows Admin Center and PowerShell remoting. Remote access should be encrypted, authenticated, restricted and logged.

6.1 Secure Shell (SSH)

ssh username@server-ip

ssh username@hostname

A successful **SSH** connection requires network reachability, a listening SSH service, appropriate firewall rules, a valid user account and correct authentication. Public-key authentication is generally preferable to password-only access for administrative environments when it is correctly managed.

6.2 Remote Administration Security Principles

- Restrict remote administration to authorized administrators and approved networks.
- Use strong authentication, preferably key-based SSH or multifactor-supported enterprise solutions.
- Avoid direct remote login as the root account where possible; use accountable administrator identities and privilege elevation.
- Log remote sessions and review authentication failures.
- Keep remote-access software patched.
- Use firewalls or access-control lists to limit management ports.
- Do not expose management interfaces unnecessarily to the public Internet.

7.0 Basic Server Readiness

A server is ready for production only when the administrator has verified that it can perform its intended role reliably and securely. Readiness is broader than simply seeing that the machine boots.

Readiness Area	Questions to Verify	Example Evidence
Identity	Is hostname correct? Is time/date/time-zone correct?	hostnamectl, timedatectl
Resources	Are CPU, RAM and storage sufficient and healthy?	lscpu, free -h, df -hT, lsblk -f
Network	Does the server have correct IP, route and DNS settings?	ip -br address, ip route, resolvectl status
Packages	Are required packages installed and repositories current?	apt list --installed, sudo apt update
Services	Are required services running and unwanted services disabled?	systemctl status, systemctl --failed
Ports	Are intended ports listening and unnecessary ports closed?	ss -tulpn
Security	Are firewall, accounts and privileges appropriate?	ufw status, id, sudo -l, permissions
Updates	Are security and stability updates applied according to policy?	apt list --upgradable, update records
Logging	Are system and service logs available for diagnosis?	journalctl, application logs
Backup / recovery	Is important configuration/data backed up and restoration understood?	backup job evidence, restore test record
Documentation	Is the server role, owner, IP, dependencies and change history recorded?	inventory/CMDB, change record

8.0 Week 3 Individual Practical: Managing Servers and Workstations

The practical exercises below are designed for a Linux virtual machine. Students should capture evidence of commands and write a short administrator-level interpretation rather than merely pasting screenshots. Commands that modify the system should be executed only in the approved laboratory environment.

8.1 Laboratory Preparation

1. Start the approved Linux VM and sign in using your student administrative account.
2. Open a terminal window.
3. Create a Week 3 evidence directory using mkdir -p ~/BSIT401/week3.
4. Create a text document or worksheet for recording command, output summary, interpretation and action.
5. Before making any change, record the current state so that you can compare before and after.

```
mkdir -p ~/BSIT401/week3
cd ~/BSIT401/week3
date
hostnamectl
```

8.2 Practical Exercise 1: Inventory Running Services

Purpose: Identify currently running system services and recognise the difference between a service and an ordinary foreground program.

Commands:

```
systemctl list-units --type=service --state=running --no-pager
```

Step-by-step procedure:

1. Run the command exactly as shown.
2. Scroll through the result and identify at least five services.
3. For each selected service, record the unit name, loaded state, active state and a short description.
4. Classify each selected service as local system support, networking, logging, scheduling, remote administration or another relevant category.
5. Choose one service that you believe is important for server operation and explain why.

Evidence to record: A screenshot or copied output showing the service list, plus a table containing five selected services and their interpretations.

How to interpret the output: Running services appear with active/running states. Their names often end in .service. A system can be operational even if some optional services are not running, so interpretation depends on the server role.

Safety / administrative caution: Do not stop or disable services during this exercise unless specifically instructed by the lecturer.

8.3 Practical Exercise 2: Identify Failed Services

Purpose: Determine whether systemd reports failed units and collect evidence before attempting any correction.

Commands:

```
systemctl --failed --no-pager
```

Step-by-step procedure:

1. Run the command.
2. If no failed units are listed, record that result and explain what it means.
3. If a failed service is listed, record its exact unit name.
4. Do not immediately restart it; first inspect its detailed status using systemctl status SERVICE --no-pager.
5. Record the key error or status message that may explain the failure.

Evidence to record: Output of systemctl --failed and, where applicable, status output for one failed service.

How to interpret the output: No failed units is evidence that systemd is not currently tracking a failed unit; it does not prove that every application-level function is healthy.

Safety / administrative caution: A failed service may be intentionally disabled or irrelevant in some environments. Diagnose before changing.

8.4 Practical Exercise 3: Inspect a Service in Detail

Purpose: Read service state, PID and recent systemd status information for a selected service.

Commands:

```
systemctl status SERVICE --no-pager
```

Step-by-step procedure:

1. Select a service that exists on your VM, such as ssh/sshd, cron, systemd-timesyncd or another lecturer-approved service.
2. Run systemctl status with the exact service name.
3. Record whether the service is loaded, active, inactive or failed.
4. Identify the main PID if one is shown.
5. Read the most recent status/log lines and summarise them in your own words.
6. State whether the service appears ready for its intended purpose.

Evidence to record: Service status output and a 4-6 sentence interpretation.

How to interpret the output: Active/running usually indicates that the main service process is operating, but readiness should also be verified using port and functional tests where relevant.

Safety / administrative caution: Do not assume that an active service is reachable from the network; verify ports and network access separately.

8.5 Practical Exercise 4: Find High-CPU Processes

Purpose: Identify processes currently using the largest share of CPU time.

Commands:

```
ps aux --sort=-%cpu | head
```

Step-by-step procedure:

1. Run the command.
2. Identify the first non-header process in the list.
3. Record USER, PID, %CPU, %MEM and COMMAND.
4. Repeat the command after 30-60 seconds.
5. Compare whether the same process remains at the top.
6. Explain why one snapshot may not be enough to diagnose a sustained CPU problem.

Evidence to record: Two command outputs and a short comparison of the top CPU-consuming process.

How to interpret the output: CPU usage can change quickly. Persistent high CPU across repeated samples is more meaningful than a single brief spike.

Safety / administrative caution: Do not kill a process merely because it appears at the top; first determine its role and whether the usage is expected.

8.6 Practical Exercise 5: Find High-Memory Processes

Purpose: Identify processes consuming the largest percentage of system memory.

Commands:

```
ps aux --sort=-%mem | head
```

Step-by-step procedure:

1. Run the command.
2. Record the top five processes by %MEM.
3. Identify whether the top process belongs to the operating system, a user application or a service.
4. Compare its PID with any service status evidence obtained earlier.
5. Explain what additional memory evidence you would collect before concluding that the system has memory pressure.

Evidence to record: Output and a table of the top five processes with a short interpretation.

How to interpret the output: A process can legitimately use a large amount of memory. Combine process data with free -h, swap activity and application behaviour before deciding there is a problem.

Safety / administrative caution: Avoid terminating unknown system processes in the lab.

8.7 Practical Exercise 6: Understand Process Relationships

Purpose: Examine parent-child process relationships and identify how services may launch worker processes.

Commands:

```
pstree -p | head -n 30
```

Step-by-step procedure:

1. Run the command.
2. Locate the init/systemd process near the top of the tree.
3. Identify at least three child processes.
4. Find one process that has its own child process if available.
5. Record the PID values shown in brackets.
6. Explain how a process tree can help diagnose a service that launches multiple workers.

Evidence to record: A screenshot or copied process tree with at least four labelled process relationships.

How to interpret the output: The process tree helps show ownership and ancestry. If a parent service terminates, its child processes may also be affected depending on how they are managed.

Safety / administrative caution: Use process relationships as evidence; do not infer service failure from ancestry alone.

8.8 Practical Exercise 7: Inspect Listening Ports

Purpose: Determine which network services are listening for incoming connections.

Commands:

```
ss -tulpn
```

Step-by-step procedure:

1. Run the command. If process details are restricted, retry with sudo only if permitted.
2. Identify at least three listening sockets, where available.
3. Record protocol, local address and port.
4. Determine whether the listening address is localhost-only, a specific interface or all interfaces.
5. Relate one listening port to a known service.
6. Explain the security significance of unnecessary listening ports.

Evidence to record: Output with three listening sockets highlighted and interpreted.

How to interpret the output: A listening socket indicates that a process is waiting for network traffic. Binding to 0.0.0.0 or :: can expose the service on multiple interfaces, subject to firewall and routing rules.

Safety / administrative caution: Do not change firewall or service bindings in this exercise unless the lecturer authorises it.

8.9 Practical Exercise 8: Refresh Package Metadata

Purpose: Use APT to refresh knowledge of available package versions without blindly changing installed software.

Commands:

```
sudo apt update
apt list --upgradable
```

Step-by-step procedure:

1. Run sudo apt update and allow it to complete.
2. Observe which repositories are contacted.
3. Record whether the command completed successfully or displayed repository/network errors.
4. Run apt list --upgradable.
5. Record the number and examples of packages reported as upgradable.
6. Explain the difference between refreshing package metadata and actually installing updates.

Evidence to record: APT update result and a summary of available upgrades.

How to interpret the output: apt update downloads repository metadata. It normally does not install the listed upgrades. apt list --upgradable shows what may be updated next.

Safety / administrative caution: Do not run a full distribution upgrade unless specifically instructed. Updates can change service behaviour and should be managed through change control.

8.10 Practical Exercise 9: Evaluate Remote-Administration Readiness

Purpose: Determine whether the VM appears ready for remote command-line administration.

Commands:

```
systemctl status ssh --no-pager
ss -tulpn | grep -E ":22\b|ssh"
ip -br address
```

Step-by-step procedure:

1. Check whether the SSH service exists and whether it is active.
2. Check whether TCP port 22 is listening.
3. Record the VM IP address.
4. If the lecturer permits a second machine or host terminal, attempt ssh username@IP_ADDRESS.
5. Record whether authentication succeeds.
6. If it fails, classify the evidence as service, port, network, firewall or authentication related.

Evidence to record: Service status, socket evidence, IP address and, where performed, connection result.

How to interpret the output: Remote administration depends on several layers. A running SSH daemon is only one requirement; network reachability, firewall policy and valid authentication are also necessary.

Safety / administrative caution: Use only authorised accounts and approved lab systems. Never test SSH against systems you do not have permission to administer.

8.11 Practical Exercise 10: Produce a Server Readiness Checklist

Purpose: Integrate Week 2 and Week 3 skills into a structured assessment of whether a Linux VM is ready for an organizational service role.

Commands:

```
hostnamectl
uptime
free -h
df -hT
systemctl --failed --no-pager
ss -tulpn
sudo apt update
```

Step-by-step procedure:

1. Record system identity and uptime.
2. Review memory and disk capacity.
3. Check for failed services.
4. Review listening ports.
5. Refresh package metadata and identify whether important updates are pending.
6. Review network identity and reachability as permitted.
7. Complete the readiness checklist in Section 7.0.
8. Assign an overall status: Ready, Ready with minor actions, or Not ready.
9. Justify the decision using at least five pieces of command evidence.

Evidence to record: A completed readiness checklist and a one-page administrator report containing evidence, risks, recommended actions and final status.

How to interpret the output: No single command proves server readiness. The decision should combine identity, capacity, service, network, patching, security and documentation evidence.

Safety / administrative caution: Do not make destructive changes while assessing readiness. Record recommendations separately from actions unless the exercise authorises correction.

9.0 Worked Administrative Examples

9.1 Example A: Web Service Reported Down

Scenario: Users report that an internal website is unavailable. systemctl status nginx shows failed and ss -tulpn shows no listener on TCP/80.

Administrator response: The evidence points first to the web service itself rather than general network routing. The administrator should inspect status and logs, identify the failure cause, correct configuration or dependency issues, restart only when safe, verify that TCP/80 is listening, and then test the web application from a client.

9.2 Example B: Service Active but Client Cannot Connect

Scenario: systemctl status ssh reports active/running, but a workstation cannot connect by SSH.

Administrator response: Because the service is active, the administrator should next verify ss -tulpn for port 22, check the server IP and route, test network reachability, inspect firewall rules, then verify authentication. This avoids incorrectly restarting a healthy service.

9.3 Example C: Update Command Fails

Scenario: sudo apt update returns a repository resolution error.

Administrator response: The administrator should not assume the package manager is broken. First verify IP configuration, default route and DNS resolution. Package-management failures can be symptoms of network or DNS problems.

9.4 Example D: Workstation Feels Slow

Scenario: ps aux --sort=-%cpu shows a user application repeatedly using high CPU, while memory and disk capacity are acceptable.

Administrator response: The evidence suggests a process-level CPU issue. The administrator should confirm persistence, identify the application owner and purpose, check logs or application state, and avoid terminating it until impact and user context are understood.

9.5 Example E: Server Has Pending Updates

Scenario: apt list --upgradable shows multiple security-related packages.

Administrator response: The server may still be operational, but patch readiness is incomplete. The administrator should review change policy, backup/recovery readiness, maintenance window and expected service restart requirements before applying updates.

10.0 Group Activity: Server / Workstation Diagnosis and Readiness Checklist

Each group receives one short scenario involving a server or workstation. The group must diagnose the likely problem and prepare a concise administration checklist. The purpose is to practise evidence-driven diagnosis rather than command memorisation.

Group Task	Required Output
Identify the intended system role	State whether it is a workstation, file server, web server, database server or another role.
List symptoms	Separate user-reported symptoms from verified facts.
Select commands	Choose commands that collect service, process, socket, package and resource evidence.
Interpret evidence	Explain what each output means and how it supports or rejects a hypothesis.
Recommend correction	Propose the least disruptive safe action.
Verify	State how you would prove service restoration or workstation readiness.
Prepare checklist	Create a 10-15 item readiness/administration checklist for the scenario.

11.0 Linux-to-Windows Administrative Equivalents

Administrative Task	Linux Example	Windows Equivalent / Tool
List services	systemctl list-units --type=service	Services console, Get-Service
Find failed/stopped services	systemctl --failed	Get-Service | Where-Object Status -ne Running (interpret carefully)
Inspect a service	systemctl status SERVICE	Get-Service, sc.exe query, Event Viewer
Inspect processes	ps aux, pstree	Task Manager, Get-Process
Top resource consumers	ps --sort, top	Task Manager, Resource Monitor, Get-Process
Listening ports	ss -tulpn	Get-NetTCPConnection, netstat -ano
Package updates	apt update / upgrade	Windows Update, winget for supported application packages
Remote command line	SSH	PowerShell Remoting / OpenSSH
Graphical remote access	Desktop tools / VNC when installed	Remote Desktop (RDP)
System logs	journalctl	Event Viewer, Get-WinEvent

12.0 Administrative Best Practices for Servers and Workstations

- Know the intended role of every managed system; administration without role context leads to unnecessary changes.
- Establish a baseline before making changes so that post-change behaviour can be compared with the original state.
- Use least privilege and accountable administrator identities.
- Apply updates through controlled maintenance processes, especially on servers used by many people.
- Change one major variable at a time when troubleshooting, then verify the result.
- Keep server/workstation inventory current: hostname, IP, OS version, owner, role, location, critical services and dependencies.
- Document commands, changes, timestamps, outcomes and rollback steps.
- Do not stop, kill, purge or disable a service until you understand its role and dependencies.
- Use secure remote administration and restrict management interfaces.
- Verify that backups are useful through restoration testing rather than assuming the existence of backup files is sufficient.
- Inspect service logs and listening ports rather than relying only on a green/running status indicator.
- Remove unnecessary software and services to reduce attack surface and maintenance burden.

13.0 Review and Formative Assessment Questions

13.1 Short-Answer / Subject Questions

1. Define a server and a workstation. Explain two administrative differences between them.

2. Explain six common organisational server roles and give one example technology for each.

3. Describe the operating-system lifecycle and explain why administrators must plan for retirement as well as installation.

4. Distinguish a Linux process from a service (daemon).

5. Explain what systemctl list-units --type=service --state=running --no-pager reveals.

6. What is the purpose of systemctl --failed --no-pager, and why does an empty result not prove that every application is healthy?

7. Explain how ps aux --sort=-%cpu | head and ps aux --sort=-%mem | head support performance diagnosis.

8. What information does pstree -p provide that a simple process list may not show as clearly?

9. Explain the importance of ss -tulpn when verifying a network server.

10. Differentiate apt update from apt upgrade.

11. Explain why updates and patches should be implemented under change control on production servers.

12. Describe the dependencies required for a successful SSH connection to a Linux server.

13. Develop a 10-point server-readiness checklist for a new web server.

14. A service shows active/running but users cannot connect. Explain a systematic evidence-based troubleshooting approach.

15. Explain why documentation is a technical control, not merely an administrative formality.

13.2 Multiple-Choice Questions

1. Which description best defines a server?

A. A computer used only by one employee
B. A system that provides services/resources to other systems
C. A device that only forwards Ethernet frames
D. A storage partition

2. Which command lists running systemd services?

A. df -hT
B. systemctl list-units --type=service --state=running --no-pager
C. ip route
D. free -h

3. Which command focuses on units currently considered failed by systemd?

A. systemctl --failed --no-pager
B. ps aux
C. apt update
D. lsblk -f

4. Which command is most appropriate for detailed status of one service?

A. systemctl status SERVICE --no-pager
B. pwd
C. nproc
D. findmnt

5. Which command sorts processes by CPU usage?

A. ps aux --sort=-%cpu | head
B. ps aux --sort=-%mem | head
C. df -hT
D. uname -a

6. Which command sorts processes by memory usage?

A. top -b
B. ps aux --sort=-%mem | head
C. lscpu
D. ip -br link

7. What is the main purpose of pstree -p?

A. Show package repositories
B. Show process hierarchy with PIDs
C. Show disk partitions
D. Show DNS servers

8. Which command shows listening network sockets?

A. ss -tulpn
B. free -h
C. lsblk
D. apt show

9. sudo apt update primarily does what?

A. Installs every upgrade automatically
B. Refreshes package indexes from repositories
C. Removes unused services
D. Restarts the server

10. apt list --upgradable is used to:

A. List packages for which newer versions are available
B. List user accounts
C. List active sockets
D. List mounted filesystems

11. Which is a common Linux remote administration protocol?

A. FTP only
B. SSH
C. ARP
D. DHCP

12. Which is most closely associated with Windows graphical remote administration?

A. RDP
B. BGP
C. NTP
D. ICMP

13. A service is active but no expected port is listening. What should the administrator do next?

A. Assume the network is fine
B. Investigate service configuration and socket binding
C. Replace the CPU
D. Delete the package manager

14. Which action best reflects safe troubleshooting?

A. Change several settings at once
B. Collect evidence and test one hypothesis at a time
C. Kill the highest CPU process immediately
D. Disable the firewall permanently

15. Which is NOT part of server readiness?

A. Required services running
B. Correct network identity
C. Known update state
D. Ignoring logs

16. A package repository cannot be reached. Which should be checked first?

A. Network and DNS connectivity
B. Monitor brightness
C. Printer paper
D. Keyboard language

17. Which server role resolves names to IP addresses?

A. DNS server
B. DHCP server
C. File server
D. Print server

18. Which server role commonly assigns IP configuration dynamically?

A. Web server
B. DHCP server
C. Database server
D. Backup server

19. Why is a server change usually more controlled than a workstation change?

A. Servers have no users
B. A server change can affect many dependent users/services
C. Workstations cannot fail
D. Servers never need updates

20. Which evidence best supports whether a service is reachable over the network?

A. Service status plus listening-port and network checks
B. CPU model only
C. Disk model only
D. Wallpaper settings

21. What is a daemon?

A. A background service process
B. A disk partition
C. A network cable
D. A file extension

22. Which statement about patching is most accurate?

A. Patches may fix vulnerabilities and defects
B. Patches never require restarts
C. Patches should always be installed without testing
D. Patches are the same as backups

23. Which lifecycle stage includes decommissioning and data handling?

A. Retirement
B. Installation
C. Login
D. Boot only

24. What does a listening socket indicate?

A. A process is waiting for network traffic on an address/port
B. RAM is full
C. The disk is encrypted
D. The CPU is virtual

25. Why should administrators document rollback steps?

A. To restore a known-good state if a change causes problems
B. To increase CPU speed
C. To create usernames
D. To disable logging

26. Which is the strongest interpretation of “systemctl --failed shows none”?

A. No systemd units are currently reported failed, but other problems may still exist
B. The server is perfect
C. The network is guaranteed healthy
D. Every application is reachable

27. What should be done before killing a high-CPU process?

A. Understand its role and verify the impact
B. Delete the OS
C. Repartition the disk
D. Disable networking

28. Which factor is essential for SSH remote access?

A. Network reachability and valid authentication
B. A local printer
C. A DVD drive
D. A graphical desktop

29. Which Windows tool is commonly used to inspect running processes interactively?

A. Task Manager
B. Notepad only
C. Paint
D. Calculator

30. Which is the best final step after correcting a service problem?

A. Verify service restoration and document the action
B. Immediately leave the system
C. Delete logs
D. Disable monitoring

13.3 Applied Scenario Questions

1. A Linux web server boots normally, but users cannot open its website. Develop a step-by-step diagnosis using service, process, port, network and log evidence.

2. A workstation becomes slow after a new application is installed. Explain how you would determine whether the problem is CPU, memory, package or service related.

3. A server has 25 upgradable packages. Explain the administrative steps you would take before, during and after applying updates in a production environment.

4. SSH is reported as active, but a remote administrator receives “connection timed out.” Identify at least five possible causes and the evidence required to test each.

5. Your department is preparing a new file server. Produce a readiness checklist covering identity, storage, network, service, security, update, backup and documentation requirements.

14.2 MCQ Answer Key

Q	Ans	Q	Ans	Q	Ans
1	B	2	B	3	A
4	A	5	A	6	B
7	B	8	A	9	B
10	A	11	B	12	A
13	B	14	B	15	D
16	A	17	A	18	B
19	B	20	A	21	A
22	A	23	A	24	A
25	A	26	A	27	A
28	A	29	A	30	A

14.3 Applied Scenario Answer Framework

Scenario 1 - Web server unavailable: Check systemctl status for the web service; systemctl --failed; ps/process evidence; ss for expected port; local application test; IP/route/DNS; firewall; logs; correct identified cause; verify from client; document.

Scenario 2 - Slow workstation: Establish symptom and baseline; inspect CPU and memory process rankings; check memory and disk capacity; identify new package/service; review logs and startup/service behaviour; test one change at a time; verify responsiveness.

Scenario 3 - Pending updates: Review update list and security relevance; confirm backup/rollback and maintenance window; communicate impact; apply approved updates; monitor package output; restart services/system if required; verify service and port functionality; document versions and outcome.

Scenario 4 - SSH timeout: Possible causes: wrong IP, no route, firewall drop, port not listening, service bound to another interface, NAT/security group issue. Use ip/route/ping where permitted, ss, systemctl, firewall evidence and logs.

Scenario 5 - New file server: Checklist should cover role/owner, hostname/time, storage/file system capacity, network/DNS, file-sharing service, permissions, firewall, update status, logging/monitoring, backup/restore, remote administration, documentation and verification from a client.

15.0 Practical Evidence and Reporting Template

Field	What Student Should Record
Exercise	Exercise number and title
Date/time	When the command/test was performed
System identity	Hostname and VM/workstation role
Command	Exact command used
Key output	Only the relevant lines/values
Interpretation	What the output means administratively
Problem / risk	Any fault, exposure or update requirement identified
Action	What was changed or recommended
Verification	How success or readiness was confirmed
Evidence	Screenshot filename or copied output reference

17.0 End-of-Week Competency Checklist

- I can distinguish a server from a workstation in terms of function and administration.
- I can identify major server roles and explain their organisational purpose.
- I can describe the operating-system lifecycle.
- I can list running services and identify failed services.
- I can inspect one service in detail using systemctl status.
- I can identify high-CPU and high-memory processes.
- I can interpret a process tree.
- I can inspect listening sockets and relate ports to services.
- I can explain apt update, upgrade and package installation concepts.
- I can describe the prerequisites for secure SSH administration.
- I can apply a server-readiness checklist.
- I can document evidence, corrective action and verification professionally.

18.0 Prescribed and Supplementary Reading

Prescribed reading aligned with the course outline:

- Nemeth, E., Snyder, G., Hein, T. R., Whaley, B., & Mackin, D. UNIX and Linux System Administration Handbook (5th ed.). Chapter 2: Booting and System Management Daemons, pp. 30-64.
- Nemeth et al. Chapter 4: Process Control, pp. 90-119.
- Nemeth et al. Chapter 6: Software Installation and Management, pp. 153-181.

Supplementary course texts may include the broader reading list approved for BSIT401, especially works on modern system administration, enterprise IT practice and network administration.

End of Lectures

Thank you